3 led multitasking#

```
/
*********************************************************************************
******
*                                                uCos-II MailBox
*********************************************************************************
*
* Processor : NXP LPC2148
* Board           : Micro-A748 SI
*
*********************************************************************************/
#include <includes.h>                    /* includes all the ucos ii functionality */
#include "func.h"                        /* includes all the board functionality */


/*--------------- AAPLICATION STACKS ---------*/
OS_STK  AppTask0stk[256];       /* Create the required number of stacks need for every child task*/
OS_STK  AppTask1stk[256];
OS_STK  AppTask2stk[256];
OS_STK  AppTask3stk[256];

/*--------------- CHILDERN TRASKS --------------*/
void  AppTask0(void *);
void  AppTask1(void *);
void  AppTask2(void *);
void  AppTask3(void *);
/*********************************************************************************
*                                  main()
*
* Description : This is the standard entry point for C code.  It is assumed that your code will call
*           main() once you have performed all necessary initialization.
*
* Argument(s) : none
```

```
*
* Return(s)   : none
*********************************************************************************/

int  main (void)
{
   BSP_Init();                          /*        Initialize Board specific programs          */
   OSInit();                            /* Initialize "uC/OS-II, The Real-Time Kernel"          */

/*----------write your application here------------------*/
            OSTaskCreate(AppTask0, 0, &AppTask0stk[255], 4);
            OSTaskCreate(AppTask1, 0, &AppTask1stk[255], 5);
            OSTaskCreate(AppTask2, 0, &AppTask2stk[255], 6);
            OSTaskCreate(AppTask3, 0, &AppTask3stk[255], 7);
/*-----------------------------------------------------*/
            OSStart();                                  /* Start multitasking (i.e. give control to uC/OS-II)      */
}



void  AppTask0(void *p_arg)
{
            p_arg = p_arg;                          /*Just to avoid compiler Warning                                */
            while(1)
   {
                        LEDset(0);
                        OSTimeDlyHMSM(0,0,0,100);
                        LEDclr(0);
                        OSTimeDlyHMSM(0,0,0,100);
   }
}
```

```
void AppTask1(void *p_arg)
{
        p_arg = p_arg;
/* Just to avoid compiler Warning                    */
        while(1)
        {
                LEDset(1);
                OSTimeDlyHMSM(0,0,0,200);
                LEDclr(1);
                OSTimeDlyHMSM(0,0,0,200);
        }
}

void AppTask2(void *p_arg)
{
        p_arg = p_arg;
/* Just to avoid compiler Warning                    */
        while(1)
        {
                LEDset(2);
                OSTimeDlyHMSM(0,0,0,300);
                LEDclr(2);
                OSTimeDlyHMSM(0,0,0,300);
        }
}

void AppTask3(void *p_arg)
{
        p_arg = p_arg;
/* Just to avoid compiler Warning                    */
```

```
        while(1)
        {
                LEDset(3);
                OSTimeDlyHMSM(0,0,0,400);
                LEDclr(3);
                OSTimeDlyHMSM(0,0,0,400);
        }
}
```

#steper motormultitask#

```
/*********************************************************************************
*                                          uCos-II Multitasking Application Code
*********************************************************************************
*
* Program Name :
* Description  : This program creates total four uCos-II tasks; out of which
*            main task acts as a parent task and create three child tasks
* Processor : NXP LPC2148
* Board           : Micro-A748
*
*********************************************************************************/
#include <includes.h>
#include "func.h"
/*--------------- AAPLICATION STACKS ---------*/
/* Create the required number of stacks needed for every child task*/
static  OS_STK  AppTask0stk[256];
static  OS_STK  AppTask1stk[256];
static  OS_STK  AppTask2stk[256];

/*-------------LOCAL FUNCTION PROTOTYPES--------------*/
```

```
void InitBoard(void);

/*--------------- CHILDERN TRASKS --------------*/
static  void  AppTask0     (void       *p_arg);
static  void  AppTask1     (void       *p_arg);
static  void  AppTask2     (void       *p_arg);
/*********************************************************************************
*                                  main()
*
* Description : This is the standard entry point for C code.  It is assumed that your code will call
*             main() once you have performed all necessary initialization.
*
* Argument(s) : none
*
* Return(s)   : none
*********************************************************************************/

int  main (void)
{
   InitBoard();                                  /*initialiase the board BSP*/
    OSInit();                                  /* Initialize "uC/OS-II, The Real-Time Kernel"            */

           OSTaskCreate(AppTask0,0,&AppTask0stk[256 - 1],4 );
           OSTaskCreate(AppTask1,0,&AppTask1stk[256 - 1],5 );
           OSTaskCreate(AppTask2,0,&AppTask2stk[256 - 1],6 );

OSStart();                                    /* Start multitasking (i.e. give control to uC/OS-II)      */
}



static  void  AppTask0 (void *p_arg)
```

```
{
  p_arg = p_arg;                          /*Just to avoid compiler Warning                    */

   while(DEF_TRUE)
   {
          /* User Code Here */
          printf("LED\n\r");
          LEDclr(0);                                        /*      Clear LED0      */
          OSTimeDlyHMSM(0,0,0,500);
          LEDset(0);                                      /*      Set LED0        */
          OSTimeDlyHMSM(0,0,0,500);
   }
}



static  void  AppTask1 (void *p_arg)
{
  int i;
  p_arg = p_arg;                          /* Just to avoid compiler Warning                    */
  while(DEF_TRUE)
  {
          /* User Code Here */
          printf("LCD:");
          for(i=0;i<27;i++)
          {
                  LCD_cmd(0x01);
                  LCD_data(i+'a');
                  OSTimeDlyHMSM(0,0,15,0);
          }
  }
```

```
}



static  void  AppTask2 (void *p_arg)
{
  p_arg = p_arg;                                  /* Just to avoid compiler Warning                  */
  printf("Stepper Motor Task started : ");
   while(DEF_TRUE)
   {
          /* User Code Here */

          StepperSet(0x08);
          OSTimeDlyHMSM(0,0,0,10);
          StepperSet(0x04);
          OSTimeDlyHMSM(0,0,0,10);
          StepperSet(0x02);
          OSTimeDlyHMSM(0,0,0,10);
          StepperSet(0x01);
          OSTimeDlyHMSM(0,0,0,10);

  }
}

void InitBoard(void)
{
          BSP_IntDisAll();              /* Disable all interrupts until we are ready to accept them */
          BSP_Init();
          InitLCD();                                                    /* Initialize LCD */
          kbdInit();                                                    /* Initialize Keyboard     */
          ADCInit();                                                        /* Initialize ADC*/
          LEDInit();                                                        /* Initialize LED */
```

```
    UartInit(9600);                                    /* Initialise the UART*/
    StepperInit();
}
```